

Proof. Theorem (Chebyshev's Sum Inequality). Let $a_1 \leq a_2 \leq \dots \leq a_n$ and $b_1 \leq b_2 \leq \dots \leq b_n$ be real numbers. Then

$$n \sum_{i=1}^n a_i b_i \geq \left(\sum_{i=1}^n a_i \right) \left(\sum_{i=1}^n b_i \right).$$

1. Rearrangement Inequality. For any permutation σ of $\{1, \dots, n\}$,

$$\sum_{i=1}^n a_i b_i \geq \sum_{i=1}^n a_i b_{\sigma(i)}. \quad (1)$$

Since both sequences are non-decreasing, the pairing in the same order yields the maximal sum (classical rearrangement inequality).

2. Average over all permutations. Consider the sum of the right-hand side of (1) over all $n!$ permutations:

$$\sum_{\sigma} \sum_{i=1}^n a_i b_{\sigma(i)}.$$

By symmetry each ordered pair (a_i, b_j) appears exactly $(n-1)!$ times, whence

$$\sum_{\sigma} \sum_{i=1}^n a_i b_{\sigma(i)} = (n-1)! \sum_{i=1}^n \sum_{j=1}^n a_i b_j. \quad (2)$$

Dividing (2) by the total number $n!$ of permutations gives the average value:

$$\begin{aligned} \frac{1}{n!} \sum_{\sigma} \sum_{i=1}^n a_i b_{\sigma(i)} &= \frac{(n-1)!}{n!} \sum_{i=1}^n \sum_{j=1}^n a_i b_j \\ &= \frac{1}{n} \left(\sum_{i=1}^n a_i \right) \left(\sum_{j=1}^n b_j \right). \end{aligned} \quad (3)$$

3. Comparison of the maximal sum with the average. The left-most sum in (1) is the maximum among all permutation sums, therefore it is at least as large as the average (3). Hence

$$\sum_{i=1}^n a_i b_i \geq \frac{1}{n} \left(\sum_{i=1}^n a_i \right) \left(\sum_{j=1}^n b_j \right). \quad (4)$$

Multiplying (4) by n yields the desired Chebyshev inequality.

4. Equality condition. Equality in (1) (and hence in (3)) occurs precisely when the maximal sum equals the average of all permutation sums. If the two sequences are not constant, there exist indices $p < q$ with $a_p < a_q$ and $b_p < b_q$; swapping b_p and b_q produces a strictly smaller sum, so the maximal sum

is strictly larger than the average. Consequently equality can hold only when at least one of the sequences is constant (or when $n = 1$, which is a special case of a constant sequence). Thus

$$n \sum_{i=1}^n a_i b_i = \left(\sum_{i=1}^n a_i \right) \left(\sum_{i=1}^n b_i \right) \iff a_1 = a_2 = \cdots = a_n \text{ or } b_1 = b_2 = \cdots = b_n.$$

Hence Chebyshev's sum inequality holds for any two non-decreasing real sequences, with equality exactly in the constant-sequence cases. $\square$ $\square$